

chapter - 2(A)

1. (i)

$$\begin{vmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 1 & 2 \\ -2 & -1 & 0 & 3 \\ -3 & -2 & -3 & 0 \end{vmatrix}$$

$$= 0 - 1 \begin{vmatrix} -1 & 1 & 2 \\ -2 & 0 & 3 \\ -3 & -3 & 0 \end{vmatrix} + 2 \begin{vmatrix} -1 & 0 & 2 \\ -2 & -1 & 3 \\ -3 & -2 & 0 \end{vmatrix} - 3 \begin{vmatrix} -1 & 0 & 1 \\ -2 & -1 & 0 \\ -3 & -2 & -3 \end{vmatrix}$$

$$= -1 [-1(0+9) - 1(0+9) + 2(6-0)] + 2 [-1(0+6) - 0 + 2(4-3)] - 3 [-1(3+0) - 0 + 1(4-3)]$$

$$= -1 [-9 - 9 + 12] + 2 [-6 + 2] - 3 [-3 + 1]$$

$$= 6 - 8 + 6$$

$$= 4$$

(ii) Let, $D = \begin{vmatrix} 3 & 1 & -1 & 2 & 1 \\ 0 & 3 & 1 & 4 & 2 \\ 1 & 4 & 5 & 2 & 1 \\ 2 & 5 & 1 & 3 & 2 \\ 1 & 2 & 3 & 2 & 2 \end{vmatrix}$

$$D = \begin{vmatrix} 3 & 1 & -1 & 2 & 1 \\ 0 & 3 & 1 & 4 & 2 \\ 1 & 4 & 5 & 2 & 1 \\ 2 & 5 & 1 & 3 & 2 \\ 1 & 2 & 3 & 2 & 2 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 4 & 2 & 3 & 1 \\ -1 & -3 & 2 & 5 \\ 1 & 2 & 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 & 3 & 1 \\ 5 & -3 & 2 & 5 \\ -1 & 2 & 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 4 & 3 & 1 \\ 5 & -1 & 2 & 5 \\ -1 & 1 & 3 & 2 \end{vmatrix} - 2 \begin{vmatrix} 1 & 4 & 2 & 1 \\ 5 & -1 & -3 & 5 \\ -1 & 1 & 2 & 2 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 4 & 2 & 3 & 1 \\ -1 & -3 & 2 & 5 \\ 1 & 2 & 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 & 3 & 1 \\ 5 & -3 & 2 & 5 \\ -1 & 2 & 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 4 & 3 & 1 \\ 5 & -1 & 2 & 5 \\ -1 & 1 & 3 & 2 \end{vmatrix} - 2 \begin{vmatrix} 1 & 4 & 2 & 1 \\ 5 & -1 & -3 & 5 \\ -1 & 1 & 2 & 2 \end{vmatrix}$$

$$+1 \begin{vmatrix} 0 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 5 & -1 & -3 & 2 \\ -1 & 1 & 2 & 3 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -2 & -8 \end{vmatrix}$$

(b) 1

$$\text{Let, } D_1 = \begin{vmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ -1 & -3 & 2 & 5 \\ 1 & 2 & 3 & 2 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 2 & 3 & 1 \\ -3 & 2 & 5 \\ 2 & 3 & 2 \end{vmatrix} - 1 \begin{vmatrix} 4 & 3 & 1 \\ -1 & 2 & 5 \\ 1 & 3 & 2 \end{vmatrix} + 4 \begin{vmatrix} 4 & 2 & 1 \\ -1 & -3 & 5 \\ 1 & 2 & 2 \end{vmatrix}$$

$$- 2 \begin{vmatrix} 4 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= 3[2(4-15) - 3(-6-10) + 1(-9-4)] - 1[4(4-15) - 3(-2-5) + 1(-3-2)] + 4[4(-6-10) - 2(-2-5) + 1(-2+3)] - 2[4(-9-4) - 2(-3+2) + 3(-2+3)]$$

$$= 3[-22 + 48 - 13] - [-44 + 21 - 5] + 4[-64 + 14 + 1] - 2[-52 + 10 + 3]$$

$$= 39 + 28 - 106 + 78$$

$$= -51$$

যাকি অঙ্ক এইভাবে D_2, D_3, D_4, D_5 অঙ্ক করে
D বের করতে হবে।

(b) Sarrus diagram :

$$(i) \begin{vmatrix} -2 & 5 & 1 \\ 3 & 4 & 2 \\ 1 & -3 & 2 \end{vmatrix}$$

$$= \begin{array}{cccccc} -2 & 5 & 1 & -2 & 5 & \\ & 3 & 4 & 2 & 3 & 4 \\ & 1 & -3 & 2 & 1 & -3 \end{array}$$

$$= (-2 \cdot 4 \cdot 2) + (5 \cdot 2 \cdot 1) + (1 \cdot 3 \cdot 3) - (1 \cdot 4 \cdot 1) - (-3 \cdot 2 \cdot -2) - (2 \cdot 3 \cdot 5)$$

$$= -16 + 10 + 9 - 4 - 12 - 30$$

$$= -63$$

(ii)

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

$$= \begin{array}{cccccc} 1 & 2 & 3 & 1 & 2 & \\ & 4 & 5 & 6 & 4 & 5 \\ & 7 & 8 & 9 & 7 & 8 \end{array}$$

$$= 45 + 84 + 98 - 105 - 48 - 72$$

$$= 0$$

$$(c) \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} : \text{multiplying all rows by } (c) \quad (1)$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{vmatrix} \quad (2)$$

$$= \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix}$$

$$= (-a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2) - (a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2 + a^2 b^2 c^2) = 4a^2 b^2 c^2 \quad [\text{proved}]$$

$$2. \begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} \quad (1)$$

$$= \begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix} \quad c_1' = c_1 + c_2 + c_3$$

$$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c) \begin{vmatrix} 0 & a+b+c & 0 \\ 0 & a+b+c & a+b+c \\ 1 & a & c+a+b \end{vmatrix} \begin{matrix} r_1' = r_2 - r_1 \\ r_2' = r_2 - r_3 \end{matrix}$$

$$= 2(a+b+c) [0 + (a+b+c)^2 - 0]$$

$$= 2(a+b+c)^3 \quad [\text{proved}]$$

3. (i)

$$\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ac \\ 1 & c & c^2 - ab \end{vmatrix}$$

$$= \begin{vmatrix} 0 & b-a & b^2 - a^2 + c(b-a) \\ 0 & b-c & b^2 - c^2 + a(b-c) \\ 1 & c & c^2 - ab \end{vmatrix}$$

$$= 0 - 0 + \begin{vmatrix} b-a & (b-a)(a+b+c) \\ b-c & (b-c)(a+b+c) \end{vmatrix}$$

$$= (b-a)(b-c)(a+b+c) - (b-a)(b-c)(a+b+c)$$

$$= 0$$

[Showed]

$$(ii) \begin{vmatrix} a^2 & a^2+ab & ab \\ bc & b^2 & b^2+bc \\ ac+c^2 & ac & c^2 \end{vmatrix}$$

$$= abc \begin{vmatrix} a & a+b & b \\ b & b & b+c \\ a+c & a & c \end{vmatrix}$$

$$= abc [a(bc-ab-ac) - (a+b)(c^2-ab-ac-bc-c^2) + b(ac-ab-bc)]$$

$$= abc [abc - a^2b - a^2c + a^2b + a^2c + abc + ab^2 + abc + b^2c + abc - ab^2 - b^2c]$$

$$= 4a^2b^2c^2 \quad [\text{showed}]$$

$$4. \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \quad r_1' = r_1 + r_2 + r_3$$

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \quad [\text{showed}]$$

$$= (a+b+c) \begin{vmatrix} 0 & a+b+c & 1 \\ -a-b+c & 2b & c-a-b \\ 0 & -(a+b+c) & 1 \end{vmatrix} \begin{matrix} c_1' = c_2 - c_1 \\ c_2' = c_3 - c_2 \end{matrix}$$

$$= (a+b+c) [0 - 0 + 1 (a+b+c)^2]$$

$$= (a+b+c)^3 \quad [\text{proved}]$$

$$5. \begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ac \\ (a+b)^2 & c^2 & ab \end{vmatrix}$$

$$= \begin{vmatrix} b^2+2bc+c^2 & a^2 & bc \\ c^2+2ac+a^2 & b^2 & ac \\ a^2+2ab+b^2 & c^2 & ab \end{vmatrix}$$

$$= \begin{vmatrix} b^2+c^2 & a^2 & bc \\ c^2+a^2 & b^2 & ac \\ a^2+b^2 & c^2 & ab \end{vmatrix} + 2 \begin{vmatrix} bc & a^2 & bc \\ ac & b^2 & ac \\ ab & c^2 & ab \end{vmatrix}$$

$$= \begin{vmatrix} a^2+b^2+c^2 & a^2 & bc \\ a^2+b^2+c^2 & b^2 & ac \\ a^2+b^2+c^2 & c^2 & ab \end{vmatrix} + 2 \cdot 0 \quad [c_1 = c_3]$$

$$= (a^2+b^2+c^2) \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ac \\ 1 & c^2 & ab \end{vmatrix}$$

$$= (a^2 + b^2 + c^2) \begin{vmatrix} 0 & a^2 - b^2 & c(b-a) \\ 0 & b^2 - c^2 & a(c-b) \\ 1 & c^2 & -ab \end{vmatrix} \begin{matrix} r_1' = r_1 - r_2 \\ r_2' = r_2 - r_3 \\ \end{matrix}$$

$$= (a^2 + b^2 + c^2) \begin{vmatrix} a^2 - b^2 & c(b-a) \\ b^2 - c^2 & a(c-b) \end{vmatrix}$$

$$= (a^2 + b^2 + c^2) \begin{vmatrix} (a+b)(a-b) & -c(a-b) \\ (b-c)(b+c) & -a(b-c) \end{vmatrix}$$

$$= (a^2 + b^2 + c^2) (a-b)(b-c) \begin{vmatrix} a+b & -c \\ b+c & -a \end{vmatrix}$$

$$= (a^2 + b^2 + c^2) (a-b)(b-c) [-a^2 - ab + bc + c^2]$$

$$= (a^2 + b^2 + c^2) (a-b)(b-c) [(c-a)(c+a) + b(c-a)]$$

$$= (a^2 + b^2 + c^2) (a-b)(b-c) (c-a)(a+b+c)$$

[proved]

$$[c^2 = 1]$$

$$7. \begin{vmatrix} -bc & bc+b^2 & bc+c^2 \\ ca+a^2 & -ca & ca+c^2 \\ ab+a^2 & ab+b^2 & -ab \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} -abc & abc+ab^2 & abc+ac^2 \\ abc+a^2b & -abc & abc+bc^2 \\ abc+a^2c & abc+b^2c & -abc \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} ab^2+abc+ac^2 & abc+ab^2 & abc+ac^2 \\ a^2b+abc+bc^2 & -abc & abc+bc^2 \\ a^2c+ab+b^2c & abc+b^2c & -abc \end{vmatrix} \quad \left. \begin{array}{l} c_1' = c_1 + c_2 \\ c_2' = c_2 + c_3 \end{array} \right\}$$

$$= \frac{1}{abc} \begin{vmatrix} a^2b+abc+a^2c & ab^2+abc+b^2c & abc+ac^2+bc^2 \\ abc+a^2b & -abc & abc+bc^2 \\ abc+a^2c & abc+b^2c & -abc \end{vmatrix} \quad \left. \begin{array}{l} r_1' = r_1 + r_2 \\ r_2' = r_2 + r_3 \end{array} \right\}$$

$$= \frac{1}{abc} \begin{vmatrix} a(ab+bc+ac) & b(ab+ac+bc) & c(bc+ca+ab) \\ a(bc+ab) & -abc & c(ab+bc) \\ a(bc+ac) & b(ac+bc) & -abc \end{vmatrix}$$

$$= (ab+bc+ac) \begin{vmatrix} 1 & 1 & 1 \\ bc+ab & -ac & ab+bc \\ bc+ac & ca+bc & -ab \end{vmatrix}$$

$$= (ab+bc+ac) \begin{vmatrix} 0 & 0 & 1 \\ bc+ab+ac & ab+bc+ac & ab+bc \\ 0 & -(ab+bc+ca) & -ab \end{vmatrix} \quad \left. \begin{array}{l} c_1' = c_2 - c_1 \\ c_2' = c_3 - c_2 \end{array} \right\}$$

$$= (ab+bc+ac) (ab+bc+ac)^2$$

$$= (ab+bc+ac)^3$$

$$6. \begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$$

$$= 2 \begin{vmatrix} 2(a+b+c) & b+c & c+a \\ 2(a+b+c) & c+a & a+b \\ 2(a+b+c) & a+b & b+c \end{vmatrix}$$

$$C_1 = C_1 + C_2 + C_3$$

$$= 2 \begin{vmatrix} a+b+c & b+c & c+a \\ a+b+c & c+a & a+b \\ a+b+c & a+b & b+c \end{vmatrix}$$

$$\begin{vmatrix} c+a \\ a+b \\ b+c \end{vmatrix}$$

$$8. \begin{vmatrix} yz - x^2 & zx - y^2 & xy - z^2 \\ zx - y^2 & xy - z^2 & yz - x^2 \\ xy - z^2 & yz - x^2 & zx - y^2 \end{vmatrix} =$$

$$= \begin{vmatrix} yz - x^2 + zx - y^2 + xy - z^2 & zx - y^2 & xy - z^2 \\ zx - y^2 + xy - z^2 + yz - x^2 & xy - z^2 & yz - x^2 \\ xy - z^2 + yz - x^2 + zx - y^2 & yz - x^2 & zx - y^2 \end{vmatrix} \quad C_1' = C_1 + C_2 + C_3$$

$$= \begin{vmatrix} (yz + zx + xy) - (x^2 + y^2 + z^2) & zx - y^2 & xy - z^2 \\ (yz + zx + xy) - (x^2 + y^2 + z^2) & xy - z^2 & yz - x^2 \\ (yz + zx + xy) - (x^2 + y^2 + z^2) & yz - x^2 & zx - y^2 \end{vmatrix}$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2) \begin{vmatrix} 1 & zx - y^2 & xy - z^2 \\ 1 & xy - z^2 & yz - x^2 \\ 1 & yz - x^2 & zx - y^2 \end{vmatrix}$$

$$r_1' = r_1 - r_2, \quad r_2' = r_2 - r_3$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2) \begin{vmatrix} 0 & zx - y^2 - xy + z^2 & xy - z^2 - yz + x^2 \\ 0 & xy - z^2 - yz + x^2 & yz - x^2 - zx + y^2 \\ 1 & yz - x^2 & zx - y^2 \end{vmatrix}$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2) \begin{vmatrix} x(z - y) + (z^2 - y^2) & y(x - z) + (x^2 - z^2) \\ y(x - z) + x^2 - z^2 & z(y - x) + y^2 - x^2 \end{vmatrix}$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2) \begin{vmatrix} (z - y)(x + z + y) & (x - z)(x + y + z) \\ (x - z)(x + z + y) & (y - x)(x + y + z) \end{vmatrix}$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2)(x + y + z)^2 \begin{vmatrix} z-y & x-z \\ x-z & y-x \end{vmatrix}$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2)(x + y + z)^2 [yz - y^2 - xz + xy - x^2 + xz + xz - z^2]$$

$$= (yz + zx + xy) - (x^2 + y^2 + z^2)(x + y + z)^2 [(yz + zx + xy) - (x^2 + y^2 + z^2)]$$

$$= \{(yz + zx + xy) - (x^2 + y^2 + z^2)\} (x + y + z)^2$$

$$12. \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$$

$$\Rightarrow - \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow (1+xyz) \begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix} = 0$$

$$\Rightarrow (1+xyz) \begin{vmatrix} 0 & (x-y) & (x^3-y^3) \\ 0 & (y-z) & (y^3-z^3) \\ 1 & z & z^3 \end{vmatrix} = 0$$

$$\Rightarrow (1+xyz) \begin{vmatrix} x-y & (x-y)(x+y) \\ y-z & (y-z)(y+z) \end{vmatrix} = 0$$

$$\Rightarrow (1 + xyz)(x-y)(y-z) \begin{vmatrix} 1 & x+y \\ 1 & y+z \end{vmatrix} = 0$$

$$\Rightarrow (1 + xyz)(x-y)(y-z)(z-x) = 0$$

Since x, y and z are different,

$$x-y \neq 0, \quad y-z \neq 0, \quad z-x \neq 0$$

$$\therefore xyz + 1 = 0$$